

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Telco Customer Churn Prediction

Project Name: Telco Customer Churn Prediction

Document Type: Business Requirements Document (BRD)

Project Domain: Telecommunications

Project Phase: Business Understanding and Data Acquisition

Version: 1.0

1. Problem Statement

Telecommunication companies face the challenge of customer churn, where customers stop using their services and switch to another provider. Customer churn can result in revenue loss and increased customer acquisition costs.

The objective of this project is to analyze customer data and identify the factors associated with customer churn. The project will later develop a machine learning model to predict which customers are likely to leave the company.

2. Project Objectives

- Analyze customer data to understand churn patterns.
- Identify customer characteristics and service-related factors associated with churn.
- Develop a machine learning classification model to predict customer churn.
- Help the business identify customers who may benefit from targeted retention strategies.

3. Business Requirements

The project should:

1. Load and analyze the Telco Customer Churn dataset.
2. Examine customer demographics, subscribed services, contract types, tenure, and payment methods.
3. Identify data quality issues, including missing values and duplicates.
4. Explore relationships between customer attributes and churn.
5. Prepare the data for future machine learning experiments.
6. Evaluate future prediction models using appropriate classification metrics.

4. Dataset Description

Dataset Name: Telco Customer Churn

Dataset Source: Kaggle

<https://www.kaggle.com/datasets/blashtchar/telco-customer-churn>

The dataset contains customer information, including demographic details, account information, subscribed services, contract type, payment method, and churn status.

- **Rows:** Approximately 7,043
- **Columns:** 21
- **Target Variable:** Churn
- **Target Values:** Yes or No

The dataset will be inspected during data acquisition to verify its structure and quality.

5. Scope of the Project

In Scope

- Dataset acquisition and documentation.
- Business requirements documentation.
- Exploratory Data Analysis (EDA).
- Data quality assessment.
- Feature preparation.
- Machine learning model development and evaluation.
- Interpretation of important churn factors.

Out of Scope

- Deploying a production system for a real telecom company.
- Automatically contacting customers.
- Implementing live customer retention campaigns.
- Integrating with a telecom company's internal systems.

6. Expected Deliverables

1. Business Requirements Document (BRD).
2. Dataset source documentation.
3. GitHub project repository.
4. Exploratory Data Analysis notebook.
5. Data Quality Report.
6. Trained machine learning classification model.
7. Model evaluation and findings report.

7. Success Criteria

The project will be evaluated based on:

- Correct identification and handling of data quality issues.
- Clear and meaningful exploratory analysis.
- Reliable evaluation of the churn prediction model.
- Appropriate use of metrics such as Precision, Recall, F1-score, and ROC-AUC.
- Useful findings that can support customer retention decisions.

Specific model performance targets will be established after a baseline model has been evaluated.

8. Assumptions and Constraints

- The supplied dataset is suitable for an educational churn prediction project.
- The target variable **Churn** represents whether a customer left the service.
- The analysis is limited to the information available in the dataset.
- Model performance on this dataset may not directly represent performance on real-world telecom data.
- Data quality issues must be investigated before model training.

9. Project Workflow

1. Understand the business problem.
2. Acquire and inspect the dataset.
3. Perform exploratory data analysis.
4. Prepare and transform the data.
5. Train classification models.
6. Evaluate model performance.
7. Interpret results and document findings.

10. Conclusion

This project aims to use customer data analysis and machine learning to predict customer churn. By identifying patterns associated with customers leaving the service, the project can help a telecommunications business make more informed customer retention decisions.

The initial phase focuses on documenting the business requirements, obtaining the dataset, and establishing the project repository. Data analysis and model development will follow in subsequent phases.